Chapter 2. Probability, Likelihood and Information Theory

Random variables, maximum likelihood, MAP estimation, Bayesian models; Entropy, KL divergence, mutual information, cross-entropy

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1 Events and Conditional Probability

A sample space Ω contains the possible outcomes and events are (measurable) subsets. The axioms are nonnegativity, $P(\Omega) = 1$ and countable additivity for disjoint events. They imply

$$P(\emptyset) = 0, \qquad P(E^c) = 1 - P(E), \qquad P(E \cup F) = P(E) + P(F) - P(E \cap F),$$

monotonicity $E \subseteq F \Rightarrow P(E) \le P(F)$, and continuity along increasing or decreasing sequences of events.

Definition 2.1: conditional probability, Bayes' rule, independence

For $P(E) > 0$, $P(F \mid E) = P(F \cap E)/P(E)$. For a partition $E_1, E_2, \dots$ of Ω,

$$P(F) = \sum_i P(F \mid E_i)P(E_i), \qquad P(E_i \mid F) = \frac{P(F \mid E_i)P(E_i)}{\sum_j P(F \mid E_j)P(E_j)}.$$

E, F are independent if $P(E \cap F) = P(E)P(F)$.

Example 2.2: why the prior matters

Let $P(D) = 0.01$, $P(+ \mid D) = 0.9$, $P(+ \mid D^c) = 0.1$. Then

$$P(D \mid +) = \frac{0.9(0.01)}{0.9(0.01) + 0.1(0.99)} \approx 0.0833.$$

A sensitive test does not make the posterior large when the prior is small. Two fair coin tosses: "first head" and "second head" each have probability $1/2$ and their intersection $1/4$, so they are independent.

2 Random Variables and Distributions

A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (e.g. the number of heads in three tosses, $X(HHT) = 2$). Its CDF is $F_X(t) = P(X \le t)$.

- Discrete: probability mass $p(x) = P(X = x)$, $F_X(t) = \sum_{x \le t} p(x)$.
- Continuous: density $f \ge 0$ with $\int f = 1$, $P(a < X \le b) = \int_a^b f(u)\,du$. A density value is not a probability and may exceed one; $P(X = x) = 0$.

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \text{ or } \int g(x)f(x)\,dx, \qquad \operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Theorem 2.3: tools used throughout the course

(i) Linearity: $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$ always.

(ii) Independence: $\mathbb{E}[XY] = \mathbb{E}X\,\mathbb{E}Y$; hence $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$. Uncorrelatedness ($\operatorname{Cov} = 0$) suffices for the variance identity.

(iii) Tower property: $\mathbb{E}[\mathbb{E}[Z \mid X]] = \mathbb{E}Z$.

(iv) $\mathbb{E}[(Z - c)^2] = \operatorname{Var}Z + (\mathbb{E}Z - c)^2 \ge \operatorname{Var}Z$ for every constant c.

Proof. (iv): write $Z - c = (Z - \mathbb{E}Z) + (\mathbb{E}Z - c)$, square, and use $\mathbb{E}[Z - \mathbb{E}Z] = 0$ to kill the cross term. The other items are standard. ∎

3 Maximum Likelihood Estimation

Definition 2.4: likelihood and MLE

For iid data $x_1, \dots, x_n$ from $p(x \mid \theta)$,

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta), \qquad \hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \sum_{i=1}^{n} \log p(x_i \mid \theta).$$

The data are fixed and the parameter varies; L is not a distribution over θ. The logarithm preserves the maximizer and turns products into sums.

Example 2.5: Bernoulli, including the boundary

$x_i \in \{0, 1\}$ with $P(x_i = 1) = \theta$; let $s = \sum x_i$, $f = n - s$. Then $\log L = s \log \theta + f \log(1 - \theta)$. For $0 < s < n$,

$$\frac{d}{d\theta} \log L = \frac{s}{\theta} - \frac{f}{1 - \theta} = 0 \iff \theta = \frac{s}{n}, \qquad \frac{d^2}{d\theta^2} \log L < 0.$$

If $s = 0$, $L = (1 - \theta)^n$ is decreasing and the maximizer is 0; if $s = n$ it is 1. Both agree with s/n, but the derivative argument cannot be used there. Eight heads in ten tosses gives 0.8; two positive reviews out of two gives the extreme estimate 1.

Example 2.6: Gaussian

For $x_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$, giving $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ (biased; the unbiased version divides by $n - 1$).

4 Bayesian Inference and MAP

Definition 2.7: prior, posterior, evidence, MAP

With prior $p(\theta)$ and likelihood $p(D \mid \theta)$,

$$p(\theta \mid D) = \frac{p(D \mid \theta) p(\theta)}{p(D)}, \qquad p(D) = \int p(D \mid \theta) p(\theta) \, d\theta,$$

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \{\log p(D \mid \theta) + \log p(\theta)\}.$$

The evidence does not depend on θ. MAP is the posterior mode; the posterior mean is a different summary.

4.1 Beta–Bernoulli conjugacy

Theorem 2.8: Beta prior

If $\theta \sim \mathrm{Beta}(a, b)$, i.e. $p(\theta) = \theta^{a-1}(1 - \theta)^{b-1} / B(a, b)$ with $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$, and s

successes, f failures are observed, then

$$\theta \mid D \sim \mathrm{Beta}(a+s,\, b+f), \qquad \mathbb{E}[\theta \mid D] = \frac{a+s}{a+b+n}, \qquad \hat{\theta}_{\mathrm{MAP}} = \frac{a+s-1}{a+b+n-2}$$

(the mode formula needs both posterior parameters above one).

Proof. $p(\theta \mid D) \propto \theta^s(1-\theta)^f \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+s)-1}(1-\theta)^{(b+f)-1}$, which is the $\mathrm{Beta}(a+s, b+f)$ kernel. For $\mathrm{Beta}(\alpha,\beta)$, $\mathbb{E}\theta = B(\alpha+1,\beta)/B(\alpha,\beta) = \alpha/(\alpha+\beta)$ using $\Gamma(z+1) = z\Gamma(z)$. The mode solves $\frac{\alpha-1}{\theta} = \frac{\beta-1}{1-\theta}$, i.e. $\theta = \frac{\alpha-1}{\alpha+\beta-2}$. ■

The uniform prior is $\mathrm{Beta}(1,1)$; then the posterior mean $(s+1)/(n+2)$ is Laplace's rule of succession, and it equals the posterior predictive probability of a success, since $P(X_{\mathrm{new}} = 1 \mid D) = \int \theta\, p(\theta \mid D)\, d\theta$.

Example 2.9: the two sellers

Seller 1 has 90 positive and 10 negative reviews; seller 2 has 2 positive and 0 negative. With uniform priors

$$\theta_1 \mid D_1 \sim \mathrm{Beta}(91, 11), \qquad \theta_2 \mid D_2 \sim \mathrm{Beta}(3, 1).$$

Posterior predictive probabilities of a positive next review: $91/102 \approx 0.892$ versus $3/4 = 0.75$, so seller 1 is preferred, whereas the MLE (0.9 versus 1) prefers seller 2. A different question is which unknown rate is larger: since $\mathrm{Beta}(3,1)$ has CDF t^3,

$$P(\theta_1 > \theta_2 \mid D) = \mathbb{E}[\theta_1^3] = \frac{91 \cdot 92 \cdot 93}{102 \cdot 103 \cdot 104} \approx 0.713.$$

5 MAP as Regularization and the Bias–Variance Decomposition

5.1 Gaussian mean with a Gaussian prior (Problem Set 1, Problem 2)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$ and prior $\theta \sim N(0, \tau^2)$.

Theorem 2.10: MAP is ridge, and it shrinks

$$\hat{\theta}_{\mathrm{MAP}} = \arg\min_{\theta} \sum_{i=1}^{n} (X_i - \theta)^2 + \lambda\theta^2, \quad \lambda = \frac{1}{\tau^2}, \qquad \hat{\theta}_{\mathrm{MAP}} = a\bar{X}, \quad a = \frac{n}{n+\lambda} = \frac{n\tau^2}{n\tau^2+1}.$$

Proof. $-\log p(\theta \mid X) = \frac{1}{2}\sum(X_i - \theta)^2 + \frac{\theta^2}{2\tau^2} + \text{const}$; multiplying by 2 gives the ridge objective. Setting the derivative $-2\sum(X_i - \theta) + 2\lambda\theta$ to zero gives $(n+\lambda)\theta = n\bar{X}$. ■

5.2 Bias–variance decomposition

Theorem 2.11: MSE = bias^2 + variance

For any estimator $\hat{\theta}$ of θ, $\mathbb{E}[(\hat{\theta} - \theta)^2] = (\mathbb{E}\hat{\theta} - \theta)^2 + \mathrm{Var}(\hat{\theta})$.

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\hat{\theta}) + (\mathbb{E}\hat{\theta} - \theta)$. The second bracket is a constant and the first has mean zero, so the cross term vanishes after taking expectations. ■

For $\hat{\theta} = a\bar{X}$: bias $(a-1)\theta$, variance a^2/n, so $\mathrm{MSE}(a) = (a-1)^2\theta^2 + a^2/n$. Minimizing over a gives $a^\star = n\theta^2/(n\theta^2+1)$, which equals the MAP shrinkage factor exactly when $\tau^2 = \theta^2$. Since $a^\star < 1$, some bias always pays off in MSE relative to the MLE $a = 1$; this is the first instance of the bias–variance trade-off that reappears for ridge regression (Chapter 3) and Adam's bias correction (Chapter 6).

Remark 2.12: Laplace prior gives the lasso

With $p(\theta) \propto e^{-|\theta|/b}$ instead, $-\log$ posterior is $\frac{1}{2}\sum(X_i-\theta)^2 + \frac{1}{b}|\theta|$ and the MAP is the soft threshold $\mathrm{sign}(\bar{X})\max(|\bar{X}| - \frac{1}{nb}, 0)$ (Chapter 3).

6 Information and Entropy

The information of an outcome is $h(x) = -\log p(x)$: rare outcomes carry more information, and for independent events probabilities multiply while information adds.

Definition 2.13: entropy

For a discrete random variable with pmf p,

$$H(X) = -\sum_x p(x)\log p(x) = \mathbb{E}[-\log p(X)], \qquad 0\log 0 := 0.$$

Natural logs give nats; base 2 gives bits.

Theorem 2.14: range of entropy

If X takes at most K values, $0 \le H(X) \le \log K$. $H = 0$ iff X is deterministic; $H = \log K$ iff X is uniform.

Proof. Each term $-p\log p \ge 0$ for $p \in [0,1]$, with equality iff $p \in \{0,1\}$. For the upper bound apply Jensen to the concave log: $H = \mathbb{E}\log\frac{1}{p(X)} \le \log\mathbb{E}\frac{1}{p(X)} = \log\sum_{x:p(x)>0} 1 \le \log K$, with equality iff $1/p(X)$ is constant on the support and the support has K points. ∎

(For continuous densities the differential entropy can be negative; the statement above is for discrete variables.)

Definition 2.15: joint and conditional entropy

$$H(X,Y) = -\sum_{x,y} p(x,y)\log p(x,y),$$

$$H(Y\mid X) = -\sum_{x,y} p(x,y)\log p(y\mid x) = \sum_x p(x) H(Y\mid X=x).$$

Theorem 2.16: chain rule

$H(X,Y) = H(X) + H(Y\mid X)$.

Proof. Take $-\mathbb{E}$ of $\log p(X,Y) = \log p(X) + \log p(Y\mid X)$. ∎

7 Kullback–Leibler Divergence

Definition 2.17: KL divergence

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_p\Big[\log \frac{p(X)}{q(X)}\Big].$$

Terms with $p(x) = 0$ contribute 0; if $p(x) > 0 = q(x)$ for some x the divergence is $+\infty$.

Theorem 2.18: Gibbs' inequality

$D_{\mathrm{KL}}(p \,\|\, q) \geq 0$, with equality iff $p = q$.

Proof. Sum over $S = \{x : p(x) > 0\}$. By Jensen for the concave log,

$$-D_{\mathrm{KL}}(p \,\|\, q) = \sum_{x \in S} p(x) \log \frac{q(x)}{p(x)} \leq \log \sum_{x \in S} p(x) \frac{q(x)}{p(x)} = \log \sum_{x \in S} q(x) \leq \log 1 = 0.$$

Equality forces (i) $q(x)/p(x) = c$ constant on S (Jensen) and (ii) $\sum_S q = 1$; then $c = \sum_S q = 1$, so $q = p$. Conversely $p = q$ gives $\log 1 = 0$ term by term. ■

Example 2.19: KL is not symmetric

For Bernoulli parameters, $D(r \,\|\, s) = r \log \frac{r}{s} + (1 - r) \log \frac{1-r}{1-s}$. With $r = 0.9$, $s = 0.5$: $D(r\|s) \approx 0.368$ but $D(s\|r) \approx 0.511$ nats. KL is not a metric (no symmetry, no triangle inequality).

8 Mutual Information

Definition 2.20: mutual information

$$I(X;Y) = D_{\mathrm{KL}}(p_{XY} \,\|\, p_X p_Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

Theorem 2.21: identities

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y) \;\geq 0,$$

with $I = 0$ iff $X \perp Y$. Consequently $H(X \mid Y) \leq H(X)$ (conditioning reduces entropy).

Proof. $\log \frac{p(x,y)}{p(x)p(y)} = \log p(x \mid y) - \log p(x)$; take $\mathbb{E}$ and use $\sum_y p(x,y) = p(x)$ to get $-H(X \mid Y) + H(X)$. The last form follows from the chain rule. Nonnegativity and the equality case are Gibbs' inequality. ■

The entropy (Venn) diagram encodes $H(X,Y) = H(X \mid Y) + I(X;Y) + H(Y \mid X)$. For a fair bit X and $Y = X$, $I = 1$ bit; for independent fair bits, $I = 0$.

9 Cross-Entropy

Definition 2.22: cross-entropy

$H(p,q) = -\sum_x p(x)\log q(x) = \mathbb{E}_p[-\log q(X)]$.

Theorem 2.23: cross-entropy = entropy + KL

$H(p,q) = H(p) + D_{\mathrm{KL}}(p\,\|\,q)$. Hence for fixed p, $\arg\min_q H(p,q) = \arg\min_q D_{\mathrm{KL}}(p\,\|\,q) = p$.

Proof. $-\sum p\log q = -\sum p\log p + \sum p\log(p/q)$. The second statement uses Gibbs. ■

Remark 2.24: classification loss and soft labels

For a one-hot label $\mathbf{e}_y$, $H(\mathbf{e}_y) = 0$ and $H(\mathbf{e}_y, q) = D_{\mathrm{KL}}(\mathbf{e}_y\|q) = -\log q_y$: the classification loss. $H(p,q)$ is *linear in its first argument*, $H(\alpha p + \beta p', q) = \alpha H(p,q) + \beta H(p',q)$, because p enters only as coefficients. Therefore a soft label that averages n one-hot labels gives $H(\frac{1}{n}\sum_j \mathbf{e}_{y_j}, q) = \frac{1}{n}\sum_j(-\log q_{y_j})$: fitting the label distribution by KL is equivalent (up to the constant $H(\hat{p})$) to ordinary cross-entropy on every individual label (2024 midterm Q2).

10 Jensen–Shannon Divergence (Problem Set 1, Problem 1)

Definition 2.25: JS divergence

With $m = \frac{1}{2}(p+q)$, $D_{\mathrm{JS}}(p\,\|\,q) = \frac{1}{2}D_{\mathrm{KL}}(p\,\|\,m) + \frac{1}{2}D_{\mathrm{KL}}(q\,\|\,m)$.

Theorem 2.26: properties

(i) $D_{\mathrm{JS}}(p\|q) \geq 0$, with equality iff $p = q$.

(ii) $D_{\mathrm{JS}}(p\|q) = D_{\mathrm{JS}}(q\|p)$.

(iii) $D_{\mathrm{JS}}(p\|q) = H(m) - \frac{1}{2}H(p) - \frac{1}{2}H(q)$.

(iv) $D_{\mathrm{JS}}(p\|q) \leq \log 2$, with equality iff p and q have disjoint supports.

Proof. (i) Both KL terms are nonnegative by Gibbs, so $D_{\mathrm{JS}} \geq 0$. If $D_{\mathrm{JS}} = 0$ then $D_{\mathrm{KL}}(p\|m) = 0$, so $p = m = \frac{1}{2}(p+q)$, i.e. $p = q$; conversely $p = q$ gives $m = p$ and both terms vanish. Note that $D_{\mathrm{KL}}(p\|m) < \infty$ always, since $m \geq p/2$. (ii) m is symmetric in p, q and the definition is a symmetric sum. (iii) $D_{\mathrm{KL}}(p\|m) = -H(p) - \sum p\log m$; averaging with the q term gives $-\frac{1}{2}H(p) - \frac{1}{2}H(q) - \sum m\log m$. (iv) Substituting m,

$$D_{\mathrm{JS}} = \log 2 + \tfrac{1}{2}\sum p\log\frac{p}{p+q} + \tfrac{1}{2}\sum q\log\frac{q}{p+q} \leq \log 2,$$

since $p/(p+q) \leq 1$ and $q/(p+q) \leq 1$. Equality iff $q = 0$ wherever $p > 0$ and vice versa. ■

Remark 2.27: JS as mutual information

Let B be a fair bit and $Z \sim p$ if $B = 0$, $Z \sim q$ if $B = 1$. Then $Z \sim m$, $H(Z \mid B) = \frac{1}{2}H(p) + \frac{1}{2}H(q)$, and by (iii) $I(Z;B) = D_{\mathrm{JS}}(p\|q) \leq H(B) = \log 2$.